

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Arava Raja(192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Arava Raja

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario: A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However, the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments
- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drone Delivery (Greedy Best-First Search & A)*

vi. Greedy Best-First Search

- Selects the path with the smallest heuristic value (estimated distance).
- Ignores the actual travel cost.

Strengths

- Fast decision-making.
- Suitable for emergency situations.

Weaknesses

- May choose blocked or expensive paths.
- Does not always find the optimal route.

ii. A* Search

- Uses:
$$f(n) = g(n) + h(n)$$
- $g(n)$: Actual travel cost.
- $h(n)$: Estimated distance to goal.

A* balances both actual cost and estimated cost, making it more reliable.

iii. Impact of Heuristic Accuracy

- Accurate heuristic → Faster and optimal search.
- Poor heuristic → More nodes explored and slower search.
- Overestimating heuristic may produce non-optimal paths.

iv. Effect of Dynamic Changes

- Newly blocked roads require replanning.
- Greedy Search may repeatedly choose blocked routes.
- A* adapts better by recalculating costs.

v. Recommendation

*A Search**

Reason

- Finds optimal route.
- Considers both cost and heuristic.
- Safer and more reliable for emergency medicine delivery.

2. Smart City Traffic Optimization (Hill Climbing)

i. Why Traditional Search is Inefficient

- Huge search space.
- Constant traffic changes.
- Very high computational cost.

ii. Hill Climbing

- Starts with one solution.
- Continuously moves to a better neighboring solution.

Limitations

- Gets stuck in local maxima.
- Cannot guarantee global optimum.

iii. Problems

Local Maximum

- Better than neighbors but not best overall.

Plateau

- Neighboring states have equal value.

Ridge

- Best path requires sideways movement which hill climbing cannot perform easily.

iv. Simulated Annealing / Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.

Genetic Algorithm

- Maintains multiple solutions.
- Uses selection, crossover and mutation.

v. Recommendation

Genetic Algorithm

Reason

- Highly scalable.
- Adapts well to changing traffic.
- Produces high-quality solutions.

3. Mars Rover (Online Search Agent)

i. Why Online Search?

- Environment is unknown.
- Complete map is unavailable.
- Decisions must be made while exploring.

ii. Challenges

- Partial observability.
- Unknown obstacles.
- Dynamic environment.
- Limited sensing capability.

iii. Internal Model

The rover:

- Collects sensor data.
- Updates its map.
- Detects hazards.
- Plans safer future routes.

iv. Comparison

Online Search	Uninformed Search	Informed Search
Learns while moving	No heuristic	Uses heuristic
Suitable for unknown maps	Explores blindly	Needs prior information
Highly adaptive	Less efficient	More efficient

v. Recommendation

Online Search Agent

Reason

- Handles uncertainty.
- Continuously updates knowledge.
- Makes safe real-time decisions.

4. University Exam Timetable (CSP)

Problem Formulation

Variables

- Exams

Domains

- Available time slots
- Available exam halls

Constraints

- No overlapping exams.
- Hall capacity limits.
- Faculty availability.
- Subject precedence.
- Special accommodations.

ii. Backtracking Search

- Assigns one exam at a time.
- Checks constraints.
- If conflict occurs, backtracks and tries another assignment.

iii. Constraint Propagation (Forward Checking)

- Removes invalid future choices.
- Detects conflicts early.
- Reduces search time.

iv. Recommendation

Backtracking + Forward Checking

Reason

- Efficient.
- Reduces unnecessary search.
- Scales better for large universities.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

- AI assumes opponent plays optimally.
- Maximizes its own score.
- Minimizes opponent's advantage.

ii. Computational Challenges

- Large branching factor.
- Deep game tree.
- High memory and time requirements.

iii. Alpha-Beta Pruning

- Eliminates branches that cannot affect the final decision.
- Evaluates fewer nodes.
- Produces the same optimal result as Minimax.

iv. Evaluation Function

Example factors:

- Material advantage.
- Position strength.
- Mobility.
- Territory control.
- Safety of important pieces.

Higher evaluation score indicates a better game state.

v. Recommendation

Minimax with Alpha-Beta Pruning

Reason

- Maintains optimal decision quality.
- Greatly reduces computation.
- Meets real-time game constraints.